Digital Info Representation Assignment

For all problems: SHOW YOUR WORK! SHOW YOUR WORK!

Answers without shown work are worth no points.

1. Use Horner's method to convert the binary unsigned whole number **`11001101`** to decimal. (15 points)

2. Use Horner's method to convert the hexadecimal unsigned whole number **`BAD1`** to decimal. (15 points)

3. Use the repeated division method to convert the unsigned whole decimal **`246`** to binary. (10 points)

4. Use the repeated division method to convert the unsigned whole decimal **`50000`** to hexadecimal. (10 points)

5. Represent your answer from the previous question as binary. (10 points)

6. Assuming an 8-bit word size, convert the signed whole decimal number -123 to binary (2's complement) (10 points)

7. Represent your answer from the previous question as hexadecimal. (10 points)

8. Since 1981, each on-road vehicle sold in the US has been required to have a 17-character VIN structured and constrained as follows:

- A three-character (*alpha-numeric*) **world manufacturer identifier** (WMI). Excludes **I**, **O & Q**
- A five-character (*alpha-numeric*) **vehicle descriptor** (VD). Excludes **I**, **O** & **Q**
- A single (*numeric* or '**X**') **check digit** (CD). The digit is <u>calculated</u> from the other 16 letters/digits (and is used for detection of invalid VIN).
- A one-character (*alpha-numeric*) **model year** (MY). Excludes '**I**', '**O**', '**Q**', '**U**', '**Z**', & '**0**' (zero)
- A one-character (*alpha-numeric*) **plant code** (PC). Excludes **I**, **O** & **Q**
- A six-character (*alpha-numeric*) **serial number** (SN). Excludes **I**, **O** & **Q**

Note that any letters used will only be UPPERCASE, and the letters **I**, **O, or Q** are never use for any part of the VIN.

From the above, calculate:

a. The theoretical maximum number of distinct VINs. Note: ok to leave answer as an algebraic expression. (10 points)

b. The minimum number of bits required to represent the above. Note: ok to leave answer as an algebraic expression. (10 points)